

BRAINARD PHILEMON JAGATI

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EDUCATION

• Indian Institute of Information Technology Nagpur

Bachelor of Technology, Computer Science Engineering (Data Science and Analytics)

◦ CGPA: 8.85/10.00

2023 - 2027

Nagpur, India

EXPERIENCE

• Nvtha AI

Research Contributor

Aug 2026 - Present

(Hybrid) Singapore

- Developed and evaluated hallucination detection methods for language models, using representation-based signals to identify unsupported outputs and adapt the approach for deployment.
- Ran experiments to compare detection approaches, analyze failure cases, and improve robustness for real-world LLM workflows.

• Singapore University of Technology & Design [\[🌐\]](#)

Visiting Student

July 2026 - Present

(On-site) Singapore

- Conducting research with [Prof. Ezekiel Soremekun](#) on the Jailbreak Attacks & Defenses for Code Large Language Models.
- Researching on white-box defense mechanisms for jailbreak attacks in open-source Code LLMs by exploiting embedding-level representations and internal model activations.

• Nanyang Technological University [\[🌐\]](#)

Research Intern

May 2026 - Present

(Remote) Singapore

- Conducting research with [Prof. Erik Cambria](#) on the safety of Multimodal Large Language Models (MLLMs) for medical applications, focusing on the interaction of clinical images and textual information.
- Investigating hallucination mitigation mechanisms in medical MLLMs using RAG-based methods.

PUBLICATIONS

1. Brainard Philemon Jagati *et al.* (Sept 2026). **Semantic-Aware Logit Adjustment for Long-Tailed Recognition.** Manuscript accepted at *NewinML workshop @NeurIPS 2026*.
2. Brainard Philemon Jagati *et al.* (Feb 2026). **A Comprehensive Survey of Logit Adjustment Methods for Long-Tailed Recognition.** *Preprint*.
3. Brainard Philemon Jagati *et al.* (Jan 2026). **Class-Confidence Aware Reweighting for Long-Tailed Learning.** *Preprint*.

PROJECTS

• Detecting Textual Backdoors through Internal Latent-Space Perturbations [\[🌐\]](#)

Tools: Python, PyTorch, Transformers, NLP

- Extended the [ONION](#) defense framework with Leave-One-Out (LOO) embedding shift and early-layer analysis to detect and mitigate textual backdoor triggers in large language models.
- Built a poisoning-and-defense evaluation pipeline using clean and BadNets-style poisoned data (5% poisoning rate), with scripts for training a poisoned victim model and testing the defense on SST-2.
- Reduced attack success rate from 16.34% to 11.95% while maintaining comparable clean accuracy (89.91% to 89.84%), with the defense fully self-contained inside the victim BERT model.

SKILLS

- **Programming Languages:** C, C++ (Proficient), Python, Java
- **Libraries/Frameworks:** TensorFlow, PyTorch, Scikit-learn, OpenCV
- **Tools:** MySQL, PowerBI, Tableau, Weka, Git, VSCode, Hugging Face
- **Relevant Coursework:** Probability & Statistics, Linear Algebra, Discrete Mathematics, Artificial Intelligence, Machine Learning, Computer Vision & Deep Learning
- **Quantitative Research:** Alpha Research & Signal Development, Factor Modeling, Cross-Sectional & Time-Series Analysis, Statistical Arbitrage, Backtesting
- **Research Interests:** Neural Collapse Theory, Long-Tailed Learning, Representation Learning, LLM Safety, Trustworthiness & Privacy.

ACHIEVEMENTS

- **Expert - Codeforces** [\[Profile\]](#)
(Peak Rating - 1612)
- **4 Stars - Codechef** [\[Profile\]](#)
(Peak Rating - 1947)
- **Global Rank 70 in Codechef Starters 211 among 30,000+ participants.**
Codechef
- **Global Rank 552 in Codeforces Round 1076 among 43,000+ participants.**
Codeforces
- **Rank 46 out of 500+ participants in TrexSim Alpha Challenge**
TrexQuant

POSITION OF RESPONSIBILITY

- **Founder**
[Code O' Phile](#)
 - Hosted nationwide coding quizzes and webinars attracting over 3000 students from various colleges.
- **Core Member**
[Dotslash \(The Coding Club\) - IIIT Nagpur](#)
 - Facilitated weekly competitive programming sessions and post contest discussions for juniors.